

EXCEL

INDEX & MATCH

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INDEX

- The INDEX function in Excel is used to return the value of a cell at a specified row and column within a given range.
- The syntax is INDEX(array, row_num, [column_num])
- array – The range of cells from which to return a value.
- row_num – The row number within the array from which to retrieve a value.
- column_num (optional) – The column number within the array (if omitted, it defaults to 1).

INDEX (CONTINUES…)

E13 =INDEX(B4:G10,2,3)

	A	B	C	D	E	F	G
1							
2		KAPASA MAKASA UNIVERSITY WEBSITE - DAILY ATTACK REPORT					
3		VIRUS	WORM	TROJAN	PHISHING	SCAM	FRAUD
4	MON	9	2	1	68	200	180
5	TUE	6	91	147	72	100	69
6	WED	13	25	235	12	300	98
7	THU	21	31	213	9	88	76
8	FRI	28	71	311	90	61	23
9	SAT	53	83	429	100	330	44
10	SUN	45	120	521	250	800	53
11							
12	SEARCH FOR A DAILY ACTIVITY						
13	DAY				147		
14	ATTACK TYPE						
15	NUMBER OF AFFECTED SYSTEMS						

- Note that our target is **147**
- Based on our section **B4:G10**, **147** is in row **2** column **3**.

MATCH

- The MATCH function in Excel is used to find the position of a value in a row or column.
- It returns the relative position of the lookup value, not the actual value itself.
- The syntax is MATCH(lookup_value, lookup_array, [match_type])
- lookup_value – The value you want to find.
- lookup_array – The range of cells where you want to search.
- match_type (optional)

i. 1 (default) – Finds the largest value less than or equal to lookup_value (array must be sorted in ascending order).

ii. 0 – Finds the exact match.

iii. -1 – Finds the smallest value greater than or equal to lookup_value (array must be sorted in descending order).

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MATCH (CONTINUES…)

E14 | =MATCH("SCAM",B3:G3,0)

KAPASA MAKASA UNIVERSITY WEBSITE - DAILY ATTACK REPORT

	VIRUS	WORM	TROJAN	PHISHING	SCAM	FRAUD
MON	9	2	1	68	200	180
TUE	6	91	147	72	100	69
WED	13	25	235	12	300	98
THU	21	31	213	9	88	76
FRI	28	71	311	90	61	23
SAT	53	83	429	100	330	44
SUN	45	120	521	250	800	53

SEARCH FOR A DAILY ACTIVITY	
DAY	147
ATTACK TYPE	5
NUMBER OF AFFECTED SYSTEMS	

- Note that our target is **SCAM**
- Based on our section **B3:G3**, **SCAM** is on position number 5.

INDEX & MATCH

- The INDEX and MATCH functions are often combined to create a more flexible and powerful alternative to VLOOKUP.
- The syntax is =INDEX(return_range, MATCH(lookup_value, lookup_column, 0))
- It can be used in Fraud Prevention & Detection Technologies to;

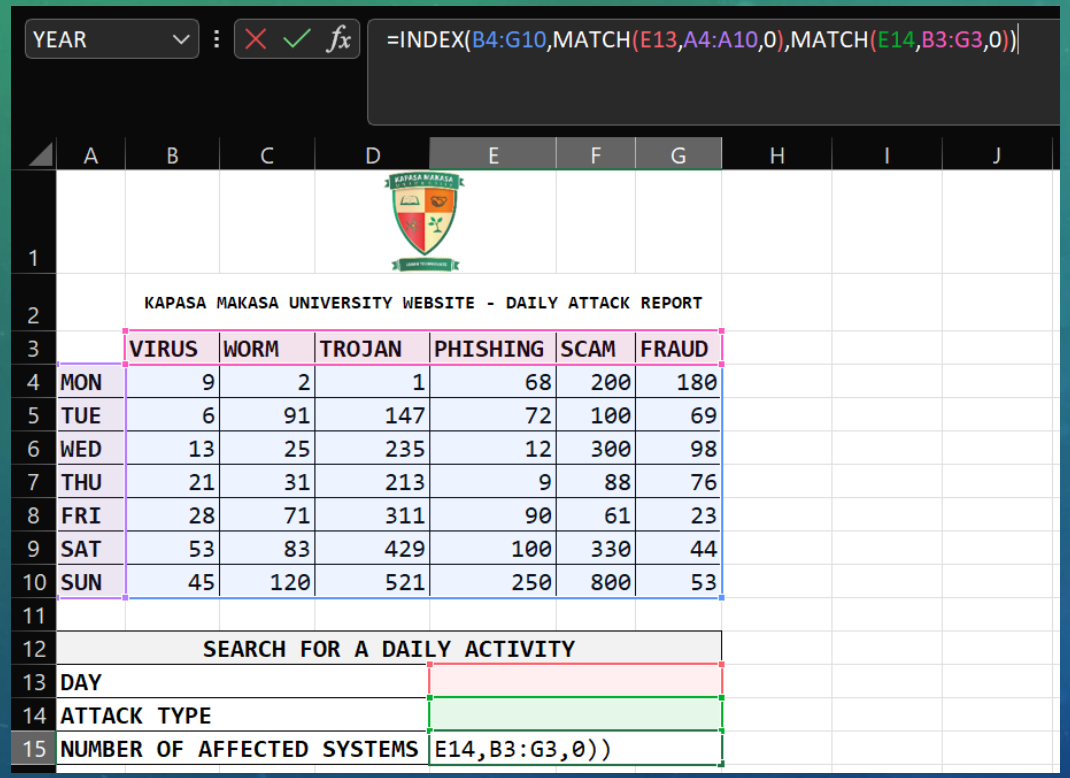
i. Detecting Duplicate Transactions

ii. Identifying Suspicious Customers

iii. Cross-Referencing Blacklisted Accounts

iv. Analyzing Fraudulent Patterns

INDEX & MATCH (CONTINUES…)



	VIRUS	WORM	TROJAN	PHISHING	SCAM	FRAUD
MON	9	2	1	68	200	180
TUE	6	91	147	72	100	69
WED	13	25	235	12	300	98
THU	21	31	213	9	88	76
FRI	28	71	311	90	61	23
SAT	53	83	429	100	330	44
SUN	45	120	521	250	800	53

- Our target array is B4:G10
- The row number is now MATCH(E13,A4:A10)
- E13 is our lookup_value to allow the user to enter the DAY between MON – SUN
- A4:A10 is our array to represent the DAY (MON – SUN) we are looking for
- The column number is now MATCH(E14,B3:G3)
- E14 is our lookup_value to allow the user to enter the ATTACK TYPE
- B3:G3 is our array to represent the ATTACK TYPE we are looking for
- 0 represents the exact match

INDEX & MATCH (CONTINUES…)

KAPASA MAKASA UNIVERSITY WEBSITE - DAILY ATTACK REPORT

	VIRUS	WORM	TROJAN	PHISHING	SCAM	FRAUD
MON	9	2	1	68	200	180
TUE	6	91	147	72	100	69
WED	13	25	235	12	300	98
THU	21	31	213	9	88	76
FRI	28	71	311	90	61	23
SAT	53	83	429	100	330	44
SUN	45	120	521	250	800	53

SEARCH FOR A DAILY ACTIVITY	
DAY	Thu
ATTACK TYPE	Worm
NUMBER OF AFFECTED SYSTEMS	31

KAPASA MAKASA UNIVERSITY WEBSITE - DAILY ATTACK REPORT

	VIRUS	WORM	TROJAN	PHISHING	SCAM	FRAUD
MON	9	2	1	68	200	180
TUE	6	91	147	72	100	69
WED	13	25	235	12	300	98
THU	21	31	213	9	88	76
FRI	28	71	311	90	61	23
SAT	53	83	429	100	330	44
SUN	45	120	521	250	800	53

SEARCH FOR A DAILY ACTIVITY	
DAY	Sun
ATTACK TYPE	Fraud
NUMBER OF AFFECTED SYSTEMS	53

ICT SOCIETY | EXCEL SLIDES

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